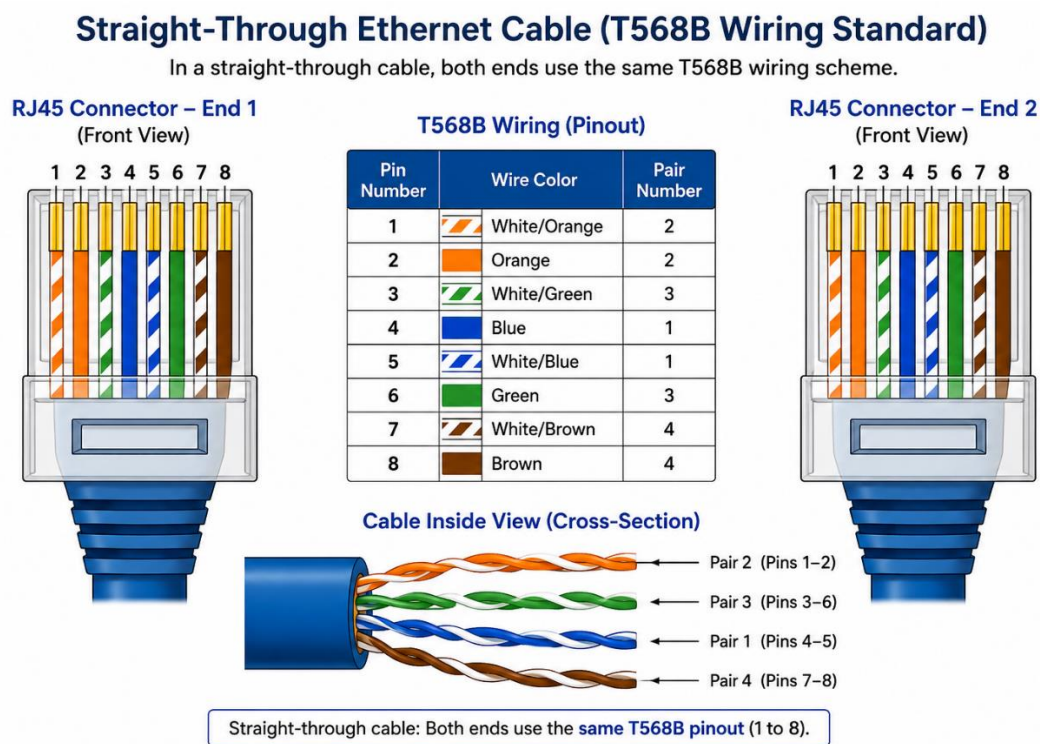


1.List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Feature	Ethernet Cable	Fiber Optic Cable
Transmission Medium	Transmits data using electrical signals through copper wires.	Transmits data using light signals through glass or plastic fibers.
Speed & Distance	Suitable for short distances (up to about 100 meters) with high speeds.	Supports much higher speeds and can transmit data over several kilometers without significant signal loss.
Cost & Installation	Less expensive and easier to install and maintain.	More expensive and requires specialized installation and equipment.

2.Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



3.Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Gather the tools: CAT6 cable, RJ45 connector, cable stripper, crimping tool, and cable tester. Strip about 2–3 cm of the cable's outer jacket. Straighten the 8 wires and arrange them in order T568B. white, orange =orange, white green blue white blue green white brown=brown=color code connect. Trim the wires so they are the same length. Insert the

wires into the RJ45 connector. Use the RJ45 crimping tool to secure the connector. Test the cable using a cable tester to make sure it works correctly.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Hint: Consider speed, distance, and budget for your answer.

For a gaming café with 10 PCs, I would use Ethernet (CAT6) cables because they are fast, reliable, and affordable. I would use an RJ45 crimping tool to attach the RJ45 connectors to the cables. Since all the PCs are in the same room, Ethernet provides a stable connection for LAN gaming. Fiber optic cables are better for long distances and are more expensive.